EFFICIENT ALGORITHMS FOR CONTEXTUAL APPLE TASTING WITH LOG-LOSS

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ABSTRACT

This paper introduces two novel algorithmic approaches designed for the contextual apple tasting problem, where the learner faces an asymmetric feedback structure by observing binary labels only upon an 'Accept' action. To address the inherent decision bias and exploration challenges, we propose two distinct but complementary strategies. First, we introduce LogCBPSide-AT, an algorithm leveraging Confidence Bounds for Partial monitoring (CBP) to explicitly quantify predictive uncertainty and effectively balance the exploration-exploitation trade-off in the logistic setting. Second, we present LogCB-AT, which reduces the apple tasting problem to online regression using an online regression oracle. This reduction-based strategy offers a computationally efficient and scalable alternative that fundamentally bypasses the complex, often intractable confidence bound constructions required by traditional methods, while accommodating substantially larger model classes. Theoretically, we prove that both algorithms achieve sublinear, instance-independent regret bounds, guaranteeing robust performance even under fundamentally restricted feedback. Moreover, LogCBPSide-AT achieves a $O(\log^2 T)$ problem-dependent regret bound, yielding stronger performance guarantees in favorable instances. The practical utility of our methods is empirically validated through adaptive Large Language Model (LLM) cascading, where they effectively optimize the trade-off between inference cost and response accuracy.

1 INTRODUCTION

In the field of sequential decision-making, the standard contextual bandit framework serves as a foundational model for learning under uncertainty. Within this setup, a learner observes a context and selects an action, receiving feedback only for the chosen larm. Achieving high performance in such tasks depends on effectively managing the balance between exploration and exploitation. The agent must decide whether to exploit its greedy action to maximize immediate rewards or explore alternative actions to gather information that enables more accurate future decision-making. While this standard framework has proven effective across a wide range of practical problems(Li et al., 2010; Tewari & Murphy, 2017; Durand et al., 2018), more challenging settings arise when the information structure is fundamentally restricted. This challenge is especially prominent in environments characterized by partial monitoring, where feedback may be entirely absent for certain choices. A particularly notable subclass of this setting is the apple tasting problem, originally introduced by Helmbold et al. (2000). In this framework, the learner faces an asymmetric information structure where the true label of an instance (e.g., whether the apple is rotten or not) is revealed only if the learner decides to 'Accept' (or 'Taste') it. If the learner chooses to remain 'Reject' (or 'Pass'), no feedback is provided, leaving the true status of the instance unknown. The learner's goal is to minimize the accumulated loss induced by the interaction between its actions (Taste vs. Pass) and the binary labels (rotten vs non-rotten), e.g., reducing the number of untasted rotten apples and tasted non-rotten apples. This objective faces a fundamental challenge where errors in rejection (false negatives) are never explicitly corrected, potentially leading to a persistent bias where the model remains stuck in a suboptimal state due to the continuous absence of corrective feedback.

In this paper, we study the contextual apple tasting problem, where action selection is informed by contextual information. The decision to Accept reveals the binary label at a specific cost, while the Reject arm does not provide any feedback, and potentially results in large loss according to the

value of y_t. This setup is motivated by modern AI application scenarios, for which we present two representative examples below:

LLM cascading The system initially processes a query through a Small Language Model (SLM) and must determine whether to cascade to a Large Language Model (LLM) for a more refined result. While the SLM provides an efficient baseline response, complex queries may require the LLM's superior reasoning at the cost of significantly higher resources and latency. Since the true performance gain is only observable upon invoking the LLM, the system operates under partial feedback: staying with the SLM prevents learning whether the LLM would have produced a better answer. Optimizing this trade-off is critical to avoid either unnecessary computational cost or degraded user satisfaction from suboptimal SLM outputs. In this example, invoking the LLM and staying with the SLM correspond to the 'Accept' and 'Reject' actions, respectively, while whether the SLM and LLM produce different answers corresponds to the binary label. The context is constructed from the SLM's internal representation before answer generation, together with its uncertainty score.

Digital Healthcare Wearable devices monitor bio-signals (contextual information) to determine if a patient needs a clinical visit. This is a quintessential contextual apple tasting problem because the actual presence of a disease (binary label) can usually only be confirmed if the device has notified the patient to see a doctor ('Accept'). If the device remains silent ('Reject'), the true medical status remains unobserved. In this context, missing an ill patient incurs a large loss, while notifying a healthy patient causes unnecessary inconvenience and medical cost.

The asymmetric feedback structure of apple tasting requires careful exploration, since labels are observed only after the revealing action. We propose two complementary approaches to this problem. First, LogCBPSide-AT adapts the confidence-bound framework of CBPSide to the logistic contextual apple tasting setting, explicitly quantifying predictive uncertainty to balance exploration and exploitation. This approach can provide strong instance-dependent guarantees, but constructing explicit confidence bounds becomes computationally demanding and difficult to extend to complex function classes. Second, we introduce LogCB-AT, a reduction-based approach that adapts SquareCB to one-sided binary feedback using an online log-loss regression oracle. By avoiding explicit confidence-bound construction, LogCB-AT accommodates broader function classes and allows straightforward integration of black-box predictive models. Both approaches exploit the apple tasting structure in which a shared latent binary label determines the losses of both actions, while feedback is observed only after acceptance.

Contributions. Our contributions are threefold. First, we propose LogCBPSide-AT, a specialization of CBPSide (Bartók & Szepesvári, 2012) to logistic contextual apple tasting. We establish a minimax-order $O(\sqrt{T})$ expected regret bound and an instance-dependent polylogarithmic regret bound under favorable problem instances. Second, we propose LogCB-AT, an oracle-efficient reduction for contextual apple tasting based on online log-loss regression, and establish an expected regret bound of $O(\sqrt{T\ \mathrm{Reg}_{\mathrm{KL}}(T)})$. Our analysis introduces a case-wise regret decomposition that aligns prediction-error terms with the revealing-action probability, enabling this guarantee for a fully randomized policy under one-sided feedback. Unlike confidence-bound methods, LogCB-AT can accommodate general function classes whenever a suitable log-loss oracle is available. Third, to the best of our knowledge, we provide the first formulation of LLM routing as a contextual apple tasting problem and demonstrate the proposed methods on adaptive LLM cascading.

2 Problem Setting and Preliminaries

We consider a contextual binary decision problem over T rounds. At each round $t \in [T] := \{1, \ldots, T\}$, the learner observes a context vector $x_t \in \mathcal{X} \subset \mathbb{R}^d$ and must choose an action $a_t \in \mathcal{A} := \{0, 1\}$, where 0 and 1 denote 'Reject' and 'Accept', respectively. The latent binary label $y_t \in \{0, 1\}$ is drawn from an unknown conditional distribution $P(\cdot \mid x_t)$. The defining feature of apple tasting is its one-sided feedback structure. Specifically, the true label y_t is revealed only when the learner chooses the revealing action ($a_t = 1$); if the learner chooses the Reject action ($a_t = 0$), no feedback is observed.

Our basic assumption is realizability: the learner has access to a function class $\mathcal{F} \subseteq (\mathcal{X} \to [0, 1])$ that contains the true conditional expectation of the binary label.

Assumption 1 (Realizability). *There exists a function $f^\star \in \mathcal{F}$ such that, for all $x \in \mathcal{X}$,*

$$f^\star(x) = \mathbb{E}[y \mid x].$$

As for LogCBPSide-AT, we further assume (in Section 4) that $\mathcal{F}$ is the class of logistic functions, whereas for LogCB-AT, we impose no additional restriction beyond the existence of a regression oracle that can efficiently learn f^* under a log-loss objective (Assumption 2 in Section 5). We denote $f^\star(x_t) = \mathbb{E}[y_t \mid x_t]$ by $y_t^\star$, the conditional mean at round t. The learner maintains an estimate $\hat{f}_{t-1}$ of $f^\star$, and its prediction at round t is $\hat{y}_t := \hat{f}_{t-1}(x_t)$. The learner is evaluated through the asymmetric loss matrix

	$y=0$	$y=1$
$a=0$	0	ℓ_{01}
$a=1$	ℓ_{10}	ℓ_{11}

(1)

where $\ell_{01}, \ell_{10}, \ell_{11} \in [0,1]$. Here, ℓ_{01} is the loss of rejecting an item with label 1, ℓ_{10} is the loss of accepting an item with label 0, and ℓ_{11} is the cost incurred when accepting an item with label 1. Throughout the paper, we assume the following cost conditions

$$\ell_{01} > \ell_{11} \geq 0 \quad \text{and} \quad \ell_{10} > 0.$$

The strict inequality $\ell_{01} > \ell_{11}$ is necessary to prevent the Reject action ($a = 0$) from trivially always being the optimal choice. This aligns naturally with practical scenarios like LLM cascading: the performance penalty of missing a better answer (ℓ_{01}) inherently outweighs the cost of invoking the LLM (ℓ_{11}). Additionally, $\ell_{11} \geq 0$ simply reflects that any invocation incurs a basic cost, while $\ell_{10} > 0$ ensures that an unnecessary call strictly incurs a computational penalty.

For any scalar $z \in [0,1]$, we define the loss function as $\psi_a(z) := \ell_{01} z(1-a) + \{\ell_{10} + (\ell_{11} - \ell_{10})z\}a$. For label y, $\psi_0(y)$ and $\psi_1(y)$ represent the losses of rejection and acceptance, respectively. Also, since $\psi_a(\cdot)$ is linear in z for each $a \in \mathcal{A}$, we have,

$$\mathbb{E}[\psi_a(y_t) \mid x_t] = \psi_a\big(f^\star(x_t)\big) = \psi_a(y_t^\star).$$

The optimal action at round t is defined as

$$a_t^\star \in \arg\min_{a \in \mathcal{A}} \psi_a(y_t^\star),$$

i.e., the action with the smaller conditional expected loss under the true model. The instantaneous regret is then defined by

$$r_t := \psi_{a_t}(y_t) - \psi_{a_t^\star}(y_t),$$

and the cumulative regret is $R(T) := \sum_{t=1}^T r_t$. Sections 4 and 5 present LogCBPSide-AT and LogCB-AT, respectively, together with their regret analyses.

3 Related work

Partial monitoring (PM) studies sequential decision problems where the learner does not observe the loss directly, but only a feedback signal determined by the chosen action and the hidden label. In stochastic finite PM, games are classically categorized into trivial, easy, hard, and hopeless classes according to their minimax regret rates (Bartók & Szepesvári, 2012; Bartók et al., 2014). Apple tasting is one of the canonical PM examples, and is known to be a two-action, two-outcome easy game, for which the minimax regret scales on the order of $\sqrt{T}$ (Heuillet et al., 2024).

Helmbold et al. (2000) was the first to formalize the apple tasting model, demonstrating that an $O(\sqrt{T})$ mistake bound is achievable through randomized algorithms in the realizable setting, where the target hypothesis is assumed to reside within the hypothesis class. Subsequently, Raman et al. (2024) extended this research from a combinatorial perspective, providing a comprehensive analysis across both realizable and agnostic settings. By introducing a new combinatorial parameter known as 'Effective Width,' they established a trichotomy for mistake bounds in the realizable setting, showing that the expected number of mistakes must fall into one of three categories: $\Theta(1)$, $\Theta(\sqrt{T})$, or $\Theta(T)$. They also characterized agnostic learnability via the Littlestone dimension. In contrast to previous literature that primarily focused on randomized algorithms, Chase & Mehalel (2024) explored the feasibility of deterministic algorithms within the realizable setting. They proved that any hypothesis

class learnable under apple tasting feedback is necessarily learnable by a deterministic algorithm. By providing an optimal mistake bound of $O(\sqrt{L(\mathcal{H})T\log T})$, where $L(\mathcal{H})$ denotes the Littlestone dimension of the hypothesis class $\mathcal{H}$.

While these works focus on non-contextual apple tasting, subsequent research expanded this framework to contextual bandits, where side information is available for decision-making. Bartók & Szepesvári (2012) extended the confidence bound based method of Bartók et al. (2012) for partial monitoring to the contextual setting, introducing CBPSide. More recently, Heuillet et al. (2024) proposed RandCBP and RandCBPSide by incorporating randomization into these frameworks to improve exploration efficiency. Kirschner et al. (2020) proposed Information-Directed Sampling (IDS) to minimize the information ratio in the linear setting. Bayesian strategies have also been adapted to better balance the exploration-exploitation trade-off in partial monitoring. Tsuchiya et al. (2020) established theoretical regret bounds for Thompson Sampling (TS), but under assumption that rewards are Gaussian. Grant & Leslie (2021) proposed PG-TS for logistic contextual apple tasting using an approximate posterior, but established sublinear regret only asymptotically. For adversarial scenarios, Harris et al. (2023) proposed an EXP-based algorithm for apple tasting.

While LogCBPSide-AT is adapted from CBPSide, it is simplified and specialized for the logistic contextual apple tasting setting. Beyond this, we establish an instance-dependent regret bound for LogCBPSide-AT, a guarantee that, to our knowledge, has not been previously derived for CBPSide. We note that the partial monitoring literature predominantly focuses on minimax bounds; among the few works that establish instance-dependent guarantees, the MED-based approach of Komiyama et al. (2015) does not prove a minimax bound, whereas LogCBPSide-AT achieves both.

Recent contextual bandit research has explored reductions to online regression oracles, beginning with SquareCB (Foster & Rakhlin, 2020). Within this line of work, Zhang et al. (2023) proposed SquareCB.G, which extends SquareCB to general feedback-graph settings. LogCB-AT differs from its apple tasting specialization in both the regression objective and the action-selection rule. While SquareCB.G uses a square-loss regression oracle, LogCB-AT employs a log-loss oracle for binary-outcome prediction. Furthermore, the apple tasting rule of SquareCB.G selects Accept deterministically whenever Accept is a greedy action, whereas LogCB-AT maintains randomization over both actions. Thus, even when equipped with the same regression oracle, the two methods retain distinct action-selection rules and differ in how predictions translate into decisions under one-sided feedback.

4 Confidence bound-based algorithm: LogCBPSide-AT

In this section, we present LogCBPSide-AT (Algorithm 1), a deterministic algorithm for the logistic contextual apple tasting problem, adapted from CBPSide Bartók & Szepesvári (2012).

We analyze the logistic function class $f_\theta(x) = \sigma(x^T\theta)$ where $\sigma(\cdot)$ is the sigmoid function. Fix $\lambda > 0$ and $S > 0$, let $\Theta := \{\theta \in \mathbb{R}^d : \|\theta\|_2 \le S\}$, and assume $\theta^* \in \Theta$. For the accepted-round index set $\mathcal{I}_t := \{s \le t : a_s = 1\}$, define the ℓ_2-regularized maximum quasi-likelihood estimator by

$$\hat{\theta}_t := \underset{\theta\in\Theta}{\arg\min} \left\{ \sum_{s\in\mathcal{I}_t} \left[\log\left(1 + e^{x_s^T\theta}\right) - y_s x_s^T\theta \right] + \frac{\lambda}{2}\|\theta\|_2^2 \right\}. \tag{2}$$

The algorithm maintains an estimate $\hat{\theta}_t$ (Line 20) for θ^* (ℓ_2-regularized maximum quasi-likelihood estimator) alongside a confidence radius $\beta_{t-1}^{x_t}(\delta_t)$ (Line 7) for the true conditional mean y_t^* given context x_t, where

$$q_n(\delta) := \left(3 + 2\log\left(1 + \frac{2}{\lambda}\right)\right) 2d\log(n+1)\log\left(\frac{d}{\delta}\right).$$

The same regularization parameter λ is used in the ridge penalty in Equation (2) and in the design matrix V_t. The algorithm then selects actions based on the relationship between this confidence interval and a decision threshold. We define the flip point $\alpha \in (0,1)$ as the value at which the expected losses of actions Accept and Reject are equal:

$$\psi_0(\alpha) = \psi_1(\alpha) \implies \alpha = \ell_{10}/(\ell_{10} + \ell_{01} - \ell_{11})$$

Algorithm 1 LogCBPSide-AT

1: **Input:** $\lambda > 0$, $S > 0$, α, $\{\delta_t\}_{t=1}^T$, c_σ
2: Observe the context x_1, Play action $a_1 = 1$ (Accept) and Observe y_1
3: $N_1 = 1$ (N_t: number of times Accept ($a = 1$) is chosen up to round t), $V_1 = \lambda I + x_1 x_1^T$
4: Compute $\hat{\theta}_1$ according to Equation 2
5: **for** $t > 1$ **do**
6: Observe the context x_t, Calculate $\hat{y}_t = \sigma(x_t^T \hat{\theta}_{t-1})$
7: Let $\beta_{t-1}^{x_t}(\delta_t) = \|x_t\|_{V_{t-1}^{-1}} \left(\frac{\sqrt{q_{N_{t-1}}(\delta_t)}}{2c_\sigma} + \frac{\sqrt{\lambda} S}{4c_\sigma} \right)$
8: **if** $|\hat{y}_t - \alpha| < \beta_{t-1}^{x_t}(\delta_t)$ **then**
9: Exploration phase
10: Play action $a_t = 1$ (Accept) and observe y_t
11: **else**
12: Exploitation phase
13: **if** $\hat{y}_t \leq \alpha$ **then**
14: Play action $a_t = 0$ (Reject)
15: **else**
16: Play action $a_t = 1$ (Accept) and observe y_t
17: **end if**
18: **end if**
19: Update $V_t = V_{t-1} + x_t x_t^T \cdot \mathbb{1}\{a_t = 1\}$, $N_t = N_{t-1} + \mathbb{1}\{a_t = 1\}$
20: Compute $\hat{\theta}_t$ according to Equation 2
21: **end for**

For any context x_t, action $a = 0$ (Reject) is the optimal action whenever $y_t^* < \alpha$ (Line 14), and action $a = 1$ (Accept) is optimal otherwise (Line 16). Accordingly, when the confidence interval lies entirely to one side of α, LogCBPSide-AT acts greedily by exploiting its current estimate $\hat{y}_t$ to select the optimal action. When the confidence interval overlaps with α, the algorithm resolves this uncertainty by selecting Accept (Lines 8-10), thereby obtaining feedback to refine future estimates and confidence bounds. We present the full procedure for LogCBPSide-AT in Algorithm 1.

The theoretical analysis of LogCBPSide-AT is deferred to Appendix C. We state the main regret guarantee below. LogCBPSide-AT achieves the minimax-optimal regret bound of order $\sqrt{T}$, consistent with the known lower bounds for easy partial monitoring games with side information.

Theorem 1 (LogCBPSide-AT minimax regret bound)**.** *Assume fixed $\lambda > 0$ and $S > 0$, $\theta^* \in \Theta$, $\sup_{x \in \mathcal{X}} \|x\|_2 \leq 1$, and $C_{\max} := \sup_{x \in \mathcal{X}, \theta \in \Theta} |x^T \theta| < \infty$. For $\delta_t = \frac{1}{t}$, the expected cumulative regret of the LogCBPSide-AT algorithm is bounded as:*

$$\mathbb{E}\left[R(T)\right] \leq O\left((1 + e^{C_{\max}})^2 e^{-C_{\max}} d\sqrt{T} \log^{\frac{3}{2}}(T)\right). \qquad \text{(Ignoring logarithmic factors in } d\text{)}$$

Furthermore, an important advantage of confidence-based algorithms in online learning with limited feedback is that they often enjoy instance dependent regret bound of order $\mathrm{polylog}(T)$. We prove that LogCBPSide-AT also enjoys a $\log^2 T$ regret bound (see Theorem 2), a guarantee that, to our knowledge, has not been previously derived for CBPSide. Proof of Theorem 2 is deferred to Appendix C.2. The instance-dependent parameter

$$\Delta \leq \min_{x \in \mathcal{X}} \left| \alpha - \sigma\left(x^\top \theta^*\right) \right|$$

provides a uniform lower bound on the separation between the true conditional probability and the decision threshold (or flip point) α. A larger value of Δ indicates greater separation from the decision boundary and, consequently, an easier problem instance.

Theorem 2 (LogCBPSide-AT instance dependent regret bound)**.** *Assume fixed $\lambda > 0$ and $S > 0$, $\theta^* \in \Theta$, $\sup_{x \in \mathcal{X}} \|x\|_2 \leq 1$, $C_{\max} := \sup_{x \in \mathcal{X}, \theta \in \Theta} |x^T \theta| < \infty$, and the existence of $\Delta > 0$ such that $\Delta \leq \min_{x \in \mathcal{X}} |\alpha - \sigma(x^T \theta^*)|$. For $\delta_t = \frac{1}{t}$, the expected cumulative regret of the LogCBPSide-AT*

algorithm is bounded as:

$$\mathbb{E}\left[R(T)\right] \leq O(\frac{1}{\Delta} d^2 \log^2 T)$$

(ignoring doubly logarithmic factors and $\mathrm{polylog}(\frac{1}{\Delta})$)

A limitation of this result, however, is its restriction to linear model classes. To address this limitation, in the next section we introduce LogCB-AT, which accommodates general function classes.

5 SQUARECB-BASED ALGORITHM: LOGCB-AT

We now introduce **LogCB-AT**, our adaptation of SquareCB to the contextual apple tasting setting. Our algorithm relies on an online log-loss regression oracle, denoted by $\mathrm{Alg}_{\mathrm{KL}}$, which produces a prediction $\hat{y}_t = \hat{f}_{t-1}(x_t)$ at round t and is updated only when the Accept action ($a_t = 1$) reveals the true label. We make the following assumption on $\mathrm{Alg}_{\mathrm{KL}}$.

Assumption 2 (Logarithmic-loss oracle guarantee)**.** *For any (possibly adaptive) sequence of contexts, actions, and labels, the online log loss regression oracle* $\mathrm{Alg}_{\mathrm{KL}}$ *produces predictions* $\hat{y}_t \in [0,1]$ *and satisfies*

$$\sum_{t=1}^{T} \mathbf{1}\{a_t = 1\}\ell_{\log}(\hat{y}_t, y_t) - \inf_{f\in\mathcal{F}} \sum_{t=1}^{T} \mathbf{1}\{a_t = 1\}\ell_{\log}\big(f(x_t), y_t\big) \leq \mathrm{Reg}_{\mathrm{KL}}(T_1) \leq \mathrm{Reg}_{\mathrm{KL}}(T),$$

where $\hat{y}_t = \hat{f}_{t-1}(x_t)$*,* $\ell_{\log}$ *denotes the standard log-loss (binary cross-entropy),*

$$\ell_{\log}(\hat{y}, y) = -\big[y \log(\hat{y}) + (1-y)\log(1-\hat{y})\big],$$

and $T_1 := \sum_{t=1}^{T} \mathbf{1}\{a_t = 1\}$ *denotes the number of rounds on which the action* $a_t = 1$ *is taken.*

Log loss is a natural objective for modeling binary outcomes through the Bernoulli likelihood. It also strongly penalizes confident errors, motivating its use in apple tasting, where opportunities to correct inaccurate predictions are limited by acceptance-only feedback. Our proposed algorithm, LogCB-AT, is presented in Algorithm 2. It closely follows the structure of SquareCB. At each round t, the learner first computes the greedy action b_t and defines the action-selection policy with a learning rate $\gamma > 0$ and an exploration parameter μ. The learner then samples an action a_t according to the policy, observes the label y_t only if $a_t = 1$, and updates $\mathrm{Alg}_{\mathrm{KL}}$ only on those rounds. A notable difference from SquareCB is that (i) the oracle uses a log-loss objective rather than a square-loss objective, and (ii) the oracle is updated only on rounds with $a_t = 1$. These differences call for non-trivial regret analysis and a new formula for the optimal value of γ which enables an expected regret bound of $O(\sqrt{T\,\mathrm{Reg}_{\mathrm{KL}}(T)})$ under mild guarantees of the oracle. We elaborate on details in the following subsections.

We analyze the regret of the LogCB-AT algorithm and derive a choice of the learning rate γ that balances the regret terms arising from one-sided binary feedback. Our analysis begins by partitioning the time steps $[T]$ into two disjoint sets, $\mathcal{I}_1$ and $\mathcal{I}_0$, according to the greedy action b_t:

$$\mathcal{I}_1 := \{t \in [T] : b_t = 1\}, \qquad \mathcal{I}_0 := \{t \in [T] : b_t = 0\}.$$

The regret is then analyzed separately over rounds in $\mathcal{I}_1$ and $\mathcal{I}_0$. Let $\bar{r}_t := \mathbb{E}_{t-1}[r_t]$ denote the conditional expected instantaneous regret, and let $\bar{R}_i(T)$ denote its cumulative sum over rounds in $\mathcal{I}_i$ for $i \in \{0,1\}$. By the tower property,

$$\mathbb{E}[R(T)] = \mathbb{E}\big[\bar{R}_1(T) + \bar{R}_0(T)\big]\,.$$

Regret for $t \in \mathcal{I}_1$. For rounds in $\mathcal{I}_1$, where Accept is the greedy action, we use the standard SquareCB decomposition and control the prediction-error term through the revealing-action probability $p_{t,1}$. By exploiting the shared latent label structure of apple tasting and converting the log-loss regret of the oracle into squared prediction error via Pinsker's inequality, we obtain

$$\mathbb{E}[\bar{R}_1(T)] \leq \frac{4T}{\gamma} + \frac{\gamma\Lambda}{8}\mathrm{Reg}_{\mathrm{KL}}(T),$$

Algorithm 2 LogCB-AT

Require: $\gamma > 0$, $\mu > 1$, online regression oracle for log loss $\mathrm{Alg}_{\mathrm{KL}}$

for $t = 1, \dots, T$ **do**
  Observe context $x_t \in \mathcal{X}$
  Obtain prediction $\hat{y}_t = \hat{f}_{t-1}(x_t)$ from $\mathrm{Alg}_{\mathrm{KL}}$
  Set $b_t \in \arg\min_{a \in \{0,1\}} \psi_a(\hat{y}_t)$
  Set

$$p_{t,a} = \frac{1}{\mu + \gamma\big(\psi_a(\hat{y}_t) - \psi_{b_t}(\hat{y}_t)\big)} \text{ for } a \neq b_t, \qquad p_{t,b_t} = 1 - \sum_{a \neq b_t} p_{t,a}$$

  Sample $a_t \sim p_t$
  if $a_t = 1$ **then**
    Observe y_t and update $\mathrm{Alg}_{\mathrm{KL}}$ with (x_t, y_t)
  end if
end for

where

$$\Lambda = (\ell_{10} - \ell_{11})^2 + \frac{\ell_{01}^2}{\mu - 1}.$$

The detailed argument is given in Appendix A.

Regret for $t \in \mathcal{I}_0$. Unlike for $t \in \mathcal{I}_1$, a direct extension of the SquareCB analysis does not yield the desired regret bound for $t \in \mathcal{I}_0$. The SquareCB decomposition contains prediction-error terms weighted by both $p_{t,1}$ and $p_{t,0}$, whereas the log-loss oracle controls only the $p_{t,1}$-weighted errors associated with the revealing action. Thus, the $p_{t,0}$-weighted terms must be related to the $p_{t,1}$-weighted ones, which introduces the probability ratio $p_{t,0}/p_{t,1}$. When Accept is greedy, this ratio is bounded by a constant independent of γ. In contrast, when Reject is greedy, the ratio grows with γ, introducing an additional dependence on γ in the direct analysis. Consequently, choosing $\gamma = \Theta\Big(\sqrt{T/\mathrm{Reg}_{\mathrm{KL}}(T)}\Big)$ leads to a suboptimal regret bound, while optimizing γ for this case yields $O\Big(\{T\mathrm{Reg}_{\mathrm{KL}}(T)\}^{2/3}\Big)$. To avoid this additional dependence, we further decompose the rounds in $\mathcal{I}_0$ according to the true optimal action $a_t^\star$. When $a_t^\star = 1$, Reject is greedy even though Accept is optimal, so the loss gap between the two actions can be controlled by the prediction error $|y_t^\star - \hat{y}_t|$. When $a_t^\star = 0$, Reject is also the optimal action, and regret occurs only when the algorithm samples Accept, naturally introducing the factor $p_{t,1}$. For $t \in \mathcal{I}_0$,

$$\bar{r}_t \leq \begin{cases} L_\Delta p_{t,1} |y_t^\star - \hat{y}_t| + \gamma L_\Delta^2 p_{t,1} (y_t^\star - \hat{y}_t)^2, & a_t^\star = 1, \\ \frac{1}{\gamma} + L_\Delta p_{t,1} |\hat{y}_t - y_t^\star|, & a_t^\star = 0, \end{cases}$$

where $L_\Delta = |\ell_{01} - \ell_{11} + \ell_{10}|$. Thus, in both cases, the prediction-error terms are weighted by the revealing probability $p_{t,1}$, avoiding the problematic ratio $p_{t,0}/p_{t,1}$ and its additional dependence on γ. Summing over the rounds in $\mathcal{I}_0$ and applying the log-loss oracle guarantee gives

$$\mathbb{E}[\bar{R}_0(T)] \leq \frac{T}{\gamma} + \frac{L_\Delta}{\sqrt{2}} \sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)} + \frac{\gamma L_\Delta^2}{2} \mathrm{Reg}_{\mathrm{KL}}(T).$$

Thus, with $\gamma = \Theta(\sqrt{T/\mathrm{Reg}_{\mathrm{KL}}(T)})$, the expected regret over rounds in $\mathcal{I}_0$ is $O(\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)})$. The detailed proof is provided in Appendix A.

Building upon the analysis above, we present the regret bound for the LogCB-AT algorithm. We set $\mu = 2$ for the two-action apple tasting setting, which ensures that the greedy action receives at least half of the sampling probability.

Theorem 3 (LogCB-AT Regret Bound)**.** *Suppose Assumptions 1 and 2 hold. For the choice of*

$$\gamma = \Theta\left(\sqrt{\frac{T}{\mathrm{Reg}_{\mathrm{KL}}(T)}}\right),$$

the expected cumulative regret of the LogCB-AT algorithm is bounded by

$$\mathbb{E}[R(T)] = O\Big(\sqrt{T \cdot \mathrm{Reg}_{\mathrm{KL}}(T)}\Big).$$

Comparison of LogCB-AT and SquareCB.G. SquareCB.G uses a square-loss regression oracle, whereas LogCB-AT uses a log-loss oracle for binary outcome prediction. Besides its natural likelihood interpretation for binary outcomes, log loss strongly penalizes confident errors, motivating its use in apple tasting, where opportunities to correct predictions are limited by acceptance-only feedback. Beyond the regression objective, the two methods differ more fundamentally in their action-selection structures. LogCB-AT is fully randomized, assigning positive probability to each action at every round, whereas SquareCB.G is partly deterministic, admitting a regime in which acceptance is selected with probability one. Thus, the exploration mechanisms of the two methods are structurally distinct, much like the distinction between randomized Thompson sampling and deterministic UCB-type methods. Prior empirical work has shown that such randomization can provide practical robustness; for example, Chapelle & Li (2011) observed that Thompson sampling is more robust than UCB to delayed feedback in the standard multi-armed bandit setting. Our experiments suggest an analogous advantage in the asymmetric-feedback setting: the fully randomized LogCB-AT achieves lower cumulative regret than SquareCB.G on all four datasets in Section 6. The relative tightness of the regret bounds for LogCB-AT and SquareCB.G depends on the loss matrix and the regret guarantees of their respective log-loss and square-loss oracles; see Appendix B for a detailed comparison.

Comparison of LogCBPSide-AT and LogCB-AT. The two algorithms provide complementary guarantees by exploiting model structure in different ways. LogCBPSide-AT explicitly leverages the logistic-linear structure and can achieve polylogarithmic regret under a positive gap condition. In contrast, LogCB-AT provides the generic oracle-based expected regret guarantee $\mathbb{E}[R(T)] = O(\sqrt{T\ \mathrm{Reg}_{\mathrm{KL}}(T)})$, so its regret adapts to the quality of the underlying log-loss oracle. For example, when equipped with an online logistic regression oracle satisfying $\mathrm{Reg}_{\mathrm{KL}}(T) = O(d \log T)$, such as ONS-GLM of Jun et al. (2017), LogCB-AT achieves $O(\sqrt{dT \log T})$ expected regret in the logistic setting. More generally, it accommodates richer hypothesis classes whenever a suitable log-loss oracle is available. Thus, LogCBPSide-AT can provide stronger instance-dependent guarantees under additional structure, whereas LogCB-AT offers greater modeling flexibility through its oracle-based formulation.

6 Experiments

Task Formulation We empirically evaluate LogCB-AT and LogCBPSide-AT on test-time adaptive Large Language Model (LLM) cascading, which we formulate as a contextual apple tasting problem, namely AppleTeA: Test-Time Adaptive LLM Cascading via Logistic Apple Tasting. At each round t, the system receives a query q_t and first generates a response using the Small Language Model (SLM), $\mathcal{M}_s$. The algorithm then chooses an action $a_t \in \{0, 1\}$: $a_t = 0$ (Reject) returns the SLM response, whereas $a_t = 1$ (Accept) invokes the stronger LLM, $\mathcal{M}_l$.

Since the ground-truth is inaccessible during test-time inference, we use the stronger LLM to construct pseudo-labels. Specifically, $y_t = 1$ if the responses of $\mathcal{M}_s$ and $\mathcal{M}_l$ differ, and $y_t = 0$ otherwise. Importantly, y_t is revealed to the routing algorithm only when $a_t = 1$, inducing the one-sided feedback structure of apple tasting.

We use the loss matrix defined in (equation 1). The loss ℓ_{10} penalizes unnecessary LLM invocation when the two models agree, while $L_{\mathrm{miss}} := \ell_{01} - \ell_{11}$ represents the additional loss incurred by retaining the SLM response when the models disagree. In our experiments, we use $\ell_{10} = 1$, $\ell_{01} = 0.6$, and $\ell_{11} = 0.1$, yielding $L_{\mathrm{miss}} = 0.5$. Further details on the models, datasets, and context construction are provided in Appendix D.

Datasets and Models. We adopt Pythia-2.8B (Biderman et al., 2023) as the SLM $\mathcal{M}_s$ and LLaMA2-13B (Touvron et al., 2023) as the strong LLM $\mathcal{M}_\ell$. We evaluate the algorithms on four benchmarks: ARC-Easy and ARC-Challenge (Clark et al., 2018), OpenBookQA (Mihaylov et al., 2018), and BoolQ (Clark et al., 2019).

Table 1: Final cumulative regret (mean ± std) over 10 random seeds.

Dataset	LogCBPSide-AT	LogCB-AT	SquareCB.G-AT	PG-TS
OpenBookQA	**71.3 ± 0.2**	77.2 ± 1.7	88.7 ± 5.3	84.3 ± 3.3
BoolQ	**472.9 ± 0.2**	515.6 ± 8.1	570.6 ± 13.1	525.8 ± 9.9
ARC-Easy	**307.0 ± 0.0**	340.6 ± 4.3	381.2 ± 7.3	356.1 ± 11.5
ARC-Challenge	**227.9 ± 0.6**	243.2 ± 2.9	267.5 ± 6.3	271.4 ± 8.8

Baselines We compare LogCB-AT and LogCBPSide-AT with two existing approaches for apple tasting: SquareCB.G-AT (Zhang et al., 2023) and Pólya–Gamma Thompson Sampling (PG-TS) (Grant & Leslie, 2021). SquareCB.G-AT specializes SquareCB.G for general feedback graphs to the apple tasting feedback structure and uses a squared-loss regression oracle. PG-TS uses logistic regression with sampling from an approximate Bayesian posterior. All methods receive the same context representation, loss matrix, and query order within each random seed.

Evaluation Metric We evaluate the algorithms using cumulative regret against the lower-loss action for each realized pseudo-label:

$$\begin{aligned}\mathrm{Regret}(t) &= \sum_{i=1}^{t} \left[L_{a_i,y_i} - \min_{a\in\{0,1\}} L_{a,y_i} \right] \\ &= \ell_{10}\,\mathrm{FP}_t + L_{\mathrm{miss}}\,\mathrm{FN}_t,\end{aligned}$$

where FP_t denotes unnecessary LLM invocations on agreement cases and FN_t denotes missed LLM invocations on disagreement cases. Pseudo-labels that are unobserved by the routing algorithm are used only for post-hoc evaluation. We note that this label-informed comparator differs from the conditional-mean-optimal policy used in our theoretical regret analysis.

Experimental Results We evaluate all methods on OpenBookQA, BoolQ, ARC-Easy, and ARC-Challenge over 10 random seeds. For each method, the hyperparameter is selected from a predefined grid according to the lowest mean final cumulative regret. Here, c_w controls the confidence width of LogCBPSide-AT, γ controls the action-sampling probabilities of LogCB-AT and SquareCB.G-AT, and M denotes the number of Gibbs samples used by PG-TS to approximate its posterior. The selected values are $c_w = 0.01$, $\gamma = 64$ for LogCB-AT, $\gamma = 64$ for SquareCB.G-AT, and $M = 15$.

Table 1 reports the final cumulative regret. Both LogCB-AT and LogCBPSide-AT consistently outperform the existing apple tasting baselines across all four datasets, with LogCBPSide-AT achieving the lowest regret overall. The relatively poor performance of SquareCB.G-AT may stem from two factors: the SquareCB.G-AT implementation used in our experiments does not explicitly exploit the shared binary outcome structure, and its action-selection rule is only partly randomized, unlike the fully randomized LogCB-AT.

7 Conclusion

In this paper, we introduced two novel algorithms, LogCBPSide-AT and LogCB-AT, designed to tackle the contextual apple tasting problem. Specifically, LogCBPSide-AT specializes the partial monitoring-based CBPSide framework for the apple tasting scenario by introducing a logistic setting. Concurrently, LogCB-AT adapts the contextual bandit-based SquareCB algorithm to operate within a log-loss setting. By leveraging these frameworks, we provided rigorous theoretical guarantees, demonstrating that our algorithms achieve sublinear regret bounds. Beyond the theoretical formulations, we empirically validated the practical utility of our approaches in a real-world application of test-time adaptive Large Language Model (LLM) cascading. Our experiments showed that dynamically routing queries between a computationally efficient Small Language Model (SLM) and a highly capable LLM via these algorithms effectively optimizes the trade-off between inference cost and response accuracy. Building on these results, future work will investigate whether the structural properties of contextual apple tasting can be further exploited to derive tighter, data-dependent regret guarantees, such as first- or second-order bounds that adapt to the difficulty of the observed sequence.

AI USE STATEMENT

We used generative AI tools to aid manuscript writing and editing, to find related work, and to assist with code implementation and debugging. The authors reviewed the AI-assisted text and code and take full responsibility for the final content and results.

REPRODUCIBILITY STATEMENT

LogCBPSide-AT and its assumptions are described in Section 4, with the complete regret proofs in Appendix C. LogCB-AT and its oracle assumptions are described in Section 5, with the complete regret proof in Appendix A. The AppleTeA evaluation procedure is given in Algorithm 3. The experimental setup is described in Section 6 and Appendix D. An implementation is provided in the supplementary material.

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Appendix

A LogCB-AT Regret Analysis

In this section, we derive an expected regret bound for the LogCB-AT algorithm. We show that the expected cumulative regret satisfies

$$\mathbb{E}[R(T)] = O\left(\sqrt{T \cdot \mathrm{Reg}_{\mathrm{KL}}(T)}\right).$$

A.1 Definition of Regret

The instantaneous regret at round t is defined as the difference between the loss of the action taken a_t and the loss of the optimal action a_t^*.

$$r_t = \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t). \tag{3}$$

The cumulative regret $R(T)$ over T rounds is the sum of these instantaneous regrets.

$$R(T) = \sum_{t=1}^{T} r_t. \tag{4}$$

Define the filtration

$$\mathcal{F}_{t-1} = \sigma\big((x_1, a_1, y_1\mathbf{1}\{a_1 = 1\}), \ldots, (x_{t-1}, a_{t-1}, y_{t-1}\mathbf{1}\{a_{t-1} = 1\}), x_t\big),$$

and let

$$\mathbb{E}_{t-1}[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}].$$

We define the conditional expected instantaneous regret as

$$\bar{r}_t := \mathbb{E}_{t-1}[r_t].$$

By realizability,

$$\mathbb{E}_{t-1}[y_t] = y_t^\star.$$

Since a_t is sampled before observing y_t and is conditionally independent of y_t given $\mathcal{F}_{t-1}$, and since $\psi_a(\cdot)$ is linear, we have

$$\begin{aligned} \bar{r}_t &= \mathbb{E}_{t-1}\left[\psi_{a_t}(y_t) - \psi_{a_t^\star}(y_t)\right] \\ &= \sum_{a \in \mathcal{A}} p_{t,a}\left(\psi_a(y_t^\star) - \psi_{a_t^\star}(y_t^\star)\right). \end{aligned} \tag{5}$$

Define

$$\bar{R}(T) := \sum_{t=1}^{T} \bar{r}_t.$$

By the tower property,

$$\mathbb{E}[R(T)] = \mathbb{E}[\bar{R}(T)]. \tag{6}$$

Lemma 1. *For any $\hat{y}, y^\star \in [0,1]$, the KL divergence between Bernoulli distributions satisfies*

$$(\hat{y} - y^\star)^2 \le \frac{1}{2} d_{\mathrm{KL}}\left(y^\star \,\|\, \hat{y}\right).$$

Proof. By Pinsker's inequality,

$$|\hat{y} - y^\star| \le \sqrt{\frac{1}{2} d_{\mathrm{KL}}\left(y^\star \,\|\, \hat{y}\right)}.$$

Squaring both sides gives

$$(\hat{y} - y^\star)^2 \le \frac{1}{2} d_{\mathrm{KL}}\left(y^\star \,\|\, \hat{y}\right).$$

□

We define the conditional expected log-loss regret on revealing rounds as

$$\overline{\mathrm{Reg}}_{\mathrm{KL}}(T) := \sum_{t=1}^{T} \mathbb{E}_{t-1}\left[\mathbf{1}\{a_t = 1\}\left(\ell_{\log}(\hat{y}_t, y_t) - \ell_{\log}(y_t^\star, y_t)\right)\right].$$

Using the conditional independence of a_t and y_t given $\mathcal{F}_{t-1}$ established above, we have

$$\begin{aligned}\overline{\mathrm{Reg}}_{\mathrm{KL}}(T) &= \sum_{t=1}^{T} p_{t,1} \mathbb{E}_{t-1}\left[\ell_{\log}(\hat{y}_t, y_t) - \ell_{\log}(y_t^\star, y_t)\right] \\ &= \sum_{t=1}^{T} p_{t,1} d_{\mathrm{KL}}\left(y_t^\star \,\|\, \hat{y}_t\right). \end{aligned} \tag{7}$$

By realizability, $f^\star \in \mathcal{F}$. Therefore, Assumption 2 implies

$$\begin{aligned}&\sum_{t=1}^{T} \mathbf{1}\{a_t = 1\}\left(\ell_{\log}(\hat{y}_t, y_t) - \ell_{\log}(y_t^\star, y_t)\right) \\ &\qquad \le \mathrm{Reg}_{\mathrm{KL}}(T).\end{aligned}$$

By the tower property,

$$\begin{aligned}\mathbb{E}\left[\overline{\mathrm{Reg}}_{\mathrm{KL}}(T)\right] &= \mathbb{E}\left[\sum_{t=1}^{T} \mathbf{1}\{a_t = 1\}\left(\ell_{\log}(\hat{y}_t, y_t) - \ell_{\log}(y_t^\star, y_t)\right)\right] \\ &\le \mathrm{Reg}_{\mathrm{KL}}(T).\end{aligned}$$

Using equation 7, we obtain

$$\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1} d_{\mathrm{KL}}\left(y_t^\star \,\|\, \hat{y}_t\right)\right] \le \mathrm{Reg}_{\mathrm{KL}}(T). \tag{8}$$

Lemma 2. *Suppose Assumptions 1 and 2 hold. Then*

$$\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2\right] \le \frac{1}{2}\mathrm{Reg}_{\mathrm{KL}}(T).$$

Proof. By Lemma 1,

$$(\hat{y}_t - y_t^\star)^2 \le \frac{1}{2} d_{\mathrm{KL}}\left(y_t^\star \,\|\, \hat{y}_t\right).$$

Therefore,

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2\right] &\le \frac{1}{2}\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1} d_{\mathrm{KL}}\left(y_t^\star \,\|\, \hat{y}_t\right)\right] \\ &\le \frac{1}{2}\mathrm{Reg}_{\mathrm{KL}}(T),\end{aligned}$$

where the last inequality follows from equation 8. □

A.2 Regret Decomposition

We partition the rounds into two disjoint sets according to the greedy action b_t:

$$\mathcal{I}_1 := \{t \in [T] : b_t = 1\}, \qquad \mathcal{I}_0 := \{t \in [T] : b_t = 0\}.$$

We further partition $\mathcal{I}_0$ according to the optimal action $a_t^\star$:

$$\mathcal{I}_{0,1} := \{t \in \mathcal{I}_0 : a_t^\star = 1\}, \qquad \mathcal{I}_{0,0} := \{t \in \mathcal{I}_0 : a_t^\star = 0\}.$$

Accordingly, define

$$\bar{R}_1(T) := \sum_{t\in\mathcal{I}_1} \bar{r}_t, \qquad \bar{R}_{0,1}(T) := \sum_{t\in\mathcal{I}_{0,1}} \bar{r}_t, \qquad \bar{R}_{0,0}(T) := \sum_{t\in\mathcal{I}_{0,0}} \bar{r}_t,$$

and

$$\bar{R}_0(T) := \bar{R}_{0,1}(T) + \bar{R}_{0,0}(T) = \sum_{t\in\mathcal{I}_{0,1}} \bar{r}_t + \sum_{t\in\mathcal{I}_{0,0}} \bar{r}_t = \sum_{t\in\mathcal{I}_0} \bar{r}_t.$$

Since $\mathcal{I}_1$ and $\mathcal{I}_0$ form a partition of $[T]$, we have

$$\bar{R}(T) = \bar{R}_1(T) + \bar{R}_0(T) = \bar{R}_1(T) + \bar{R}_{0,1}(T) + \bar{R}_{0,0}(T).$$

Therefore, by the relation $\mathbb{E}[R(T)] = \mathbb{E}[\bar{R}(T)]$ established above,

$$\mathbb{E}[R(T)] = \mathbb{E}\left[\bar{R}_1(T) + \bar{R}_0(T)\right] = \mathbb{E}\left[\bar{R}_1(T) + \bar{R}_{0,1}(T) + \bar{R}_{0,0}(T)\right].$$

Lemma 3. *Suppose Assumptions 1 and 2 hold, and let $\mu = 2$. Define*

$$L_{\mathrm{A}} := |\ell_{11} - \ell_{10}|$$

and

$$\Lambda := L_{\mathrm{A}}^2 + \frac{\ell_{01}^2}{\mu - 1}.$$

Then, the expected regret over the rounds in $\mathcal{I}_1$ is bounded by

$$\mathbb{E}\left[\bar{R}_1(T)\right] \le \frac{4T}{\gamma} + \frac{\gamma\Lambda}{8}\mathrm{Reg}_{\mathrm{KL}}(T)$$

Proof. Recall that for $t \in \mathcal{I}_1$, the greedy action is $b_t = 1$. By the definition of $\bar{R}_1(T)$,

$$\bar{R}_1(T) = \sum_{t\in\mathcal{I}_1}\sum_{a\in\mathcal{A}} p_{t,a}\left(\psi_a(y_t^\star) - \psi_{a_t^\star}(y_t^\star)\right).$$

Adding and subtracting the squared prediction-error term gives

$$\begin{aligned}\bar{R}_1(T) = &\sum_{t\in\mathcal{I}_1}\sum_{a\in\mathcal{A}} p_{t,a}\left[\psi_a(y_t^\star) - \psi_{a_t^\star}(y_t^\star) - \frac{\gamma}{4}\left(\psi_a(\hat{y}_t) - \psi_a(y_t^\star)\right)^2\right] \\ &+ \frac{\gamma}{4}\sum_{t\in\mathcal{I}_1}\sum_{a\in\mathcal{A}} p_{t,a}\left(\psi_a(\hat{y}_t) - \psi_a(y_t^\star)\right)^2.\end{aligned}$$

For $\mu = 2$, Lemma 3 of Foster & Rakhlin (2020) implies that, for every round t,

$$\sum_{a\in\mathcal{A}} p_{t,a}\left[\psi_a(y_t^\star) - \psi_{a_t^\star}(y_t^\star) - \frac{\gamma}{4}\left(\psi_a(\hat{y}_t) - \psi_a(y_t^\star)\right)^2\right] \le \frac{4}{\gamma}.$$

Therefore,

$$\begin{aligned}&\sum_{t\in\mathcal{I}_1}\sum_{a\in\mathcal{A}} p_{t,a}\left[\psi_a(y_t^\star) - \psi_{a_t^\star}(y_t^\star) - \frac{\gamma}{4}\left(\psi_a(\hat{y}_t) - \psi_a(y_t^\star)\right)^2\right] \\ &\qquad \le \frac{4|\mathcal{I}_1|}{\gamma} \le \frac{4T}{\gamma}.\end{aligned}$$

It remains to bound the squared prediction-error term. By the definition of the loss functions,

$$\left(\psi_1(\hat{y}_t) - \psi_1(y_t^\star)\right)^2 = L_{\mathrm{A}}^2(\hat{y}_t - y_t^\star)^2,$$

and

$$\left(\psi_0(\hat{y}_t) - \psi_0(y_t^\star)\right)^2 = \ell_{01}^2(\hat{y}_t - y_t^\star)^2.$$

Since $b_t = 1$ for $t \in \mathcal{I}_1$, we have

$$\psi_1(\hat{y}_t) \le \psi_0(\hat{y}_t).$$

The sampling probabilities therefore satisfy

$$p_{t,0} = \frac{1}{\mu + \gamma\left(\psi_0(\hat{y}_t) - \psi_1(\hat{y}_t)\right)}, \qquad p_{t,1} = 1 - p_{t,0}.$$

Hence,

$$\begin{aligned}\frac{p_{t,0}}{p_{t,1}} &= \frac{1}{\mu + \gamma\left(\psi_0(\hat{y}_t) - \psi_1(\hat{y}_t)\right) - 1} \\ &\le \frac{1}{\mu - 1}.\end{aligned}$$

Therefore, for every $t \in \mathcal{I}_1$,

$$\begin{aligned}&\sum_{a\in\mathcal{A}} p_{t,a}\left(\psi_a(\hat{y}_t) - \psi_a(y_t^\star)\right)^2 \\ &= p_{t,1}L_{\mathrm{A}}^2(\hat{y}_t - y_t^\star)^2 + p_{t,0}\ell_{01}^2(\hat{y}_t - y_t^\star)^2 \\ &\le p_{t,1}\left(L_{\mathrm{A}}^2 + \frac{\ell_{01}^2}{\mu - 1}\right)(\hat{y}_t - y_t^\star)^2 \\ &= \Lambda p_{t,1}(\hat{y}_t - y_t^\star)^2.\end{aligned}$$

Summing over $t \in \mathcal{I}_1$ and taking expectation gives

$$\begin{aligned}&\mathbb{E}\left[\sum_{t\in\mathcal{I}_1}\sum_{a\in\mathcal{A}} p_{t,a}\left(\psi_a(\hat{y}_t) - \psi_a(y_t^\star)\right)^2\right] \\ &\le \Lambda\mathbb{E}\left[\sum_{t\in\mathcal{I}_1} p_{t,1}(\hat{y}_t - y_t^\star)^2\right] \\ &\le \Lambda\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2\right] \\ &\le \frac{1}{2}\Lambda\mathrm{Reg}_{\mathrm{KL}}(T),\end{aligned}$$

where the last inequality follows from Lemma 2. Combining the two bounds yields

$$\mathbb{E}\left[\bar{R}_1(T)\right] \le \frac{4T}{\gamma} + \frac{\gamma\Lambda}{8}\mathrm{Reg}_{\mathrm{KL}}(T),$$

which completes the proof. □

Lemma 4. *Suppose Assumptions 1 and 2 hold, and let $\mu = 2$. Define*

$$L_\Delta := \left|\ell_{01} - \ell_{11} + \ell_{10}\right|.$$

Then, the expected regret over the rounds in $\mathcal{I}_0$ is bounded by

$$\mathbb{E}\left[\bar{R}_0(T)\right] \le \frac{T}{\gamma} + \frac{L_\Delta}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)} + \frac{\gamma L_\Delta^2}{2}\mathrm{Reg}_{\mathrm{KL}}(T).$$

Proof. For any round $t \in \mathcal{I}_0$, the greedy action is $b_t = 0$. Define

$$\Delta(y) := \psi_0(y) - \psi_1(y),$$

and let

$$\widehat{\Delta}_t := \Delta(\hat{y}_t), \qquad \Delta_t^\star := \Delta(y_t^\star).$$

Since $b_t = 0$, we have

$$\widehat{\Delta}_t \le 0.$$

Moreover, since $\Delta(\cdot)$ is affine in y,

$$\left|\Delta(y) - \Delta(y')\right| = L_\Delta |y - y'|.$$

For $\mu = 2$, the probability of choosing action 1 is

$$p_{t,1} = \frac{1}{2 + \gamma(-\widehat{\Delta}_t)}.$$

First, consider $t \in \mathcal{I}_{0,1}$, where $a_t^\star = 1$. In this case, $\Delta_t^\star \ge 0$. The conditional expected regret is

$$\begin{aligned}\bar{r}_t &= p_{t,0}\left(\psi_0(y_t^\star) - \psi_1(y_t^\star)\right) \\ &= p_{t,0}\Delta_t^\star.\end{aligned}$$

Since

$$\frac{p_{t,0}}{p_{t,1}} = 1 + \gamma(-\widehat{\Delta}_t),$$

we have

$$\bar{r}_t = p_{t,1}\left(1 + \gamma(-\widehat{\Delta}_t)\right)\Delta_t^\star.$$

Since $\widehat{\Delta}_t \le 0 \le \Delta_t^\star$,

$$-\widehat{\Delta}_t \le \Delta_t^\star - \widehat{\Delta}_t \le L_\Delta |y_t^\star - \hat{y}_t|,$$

and

$$\Delta_t^\star \le \Delta_t^\star - \widehat{\Delta}_t \le L_\Delta |y_t^\star - \hat{y}_t|.$$

Therefore,

$$\bar{r}_t \le L_\Delta p_{t,1} |y_t^\star - \hat{y}_t| + \gamma L_\Delta^2 p_{t,1} (y_t^\star - \hat{y}_t)^2.$$

Next, consider $t \in \mathcal{I}_{0,0}$, where $a_t^\star = 0$. In this case, $\Delta_t^\star \le 0$. The conditional expected regret is

$$\begin{aligned}\bar{r}_t &= p_{t,1}\left(\psi_1(y_t^\star) - \psi_0(y_t^\star)\right) \\ &= p_{t,1}(-\Delta_t^\star).\end{aligned}$$

By adding and subtracting $\widehat{\Delta}_t$,

$$\begin{aligned}-\Delta_t^\star &= -\widehat{\Delta}_t + \left(\widehat{\Delta}_t - \Delta_t^\star\right) \\ &\le -\widehat{\Delta}_t + L_\Delta |\hat{y}_t - y_t^\star|.\end{aligned}$$

Hence,

$$\bar{r}_t \le p_{t,1}(-\widehat{\Delta}_t) + L_\Delta p_{t,1} |\hat{y}_t - y_t^\star|.$$

Since

$$p_{t,1}(-\widehat{\Delta}_t) = \frac{-\widehat{\Delta}_t}{2 + \gamma(-\widehat{\Delta}_t)} \le \frac{1}{\gamma},$$

we obtain

$$\bar{r}_t \le \frac{1}{\gamma} + L_\Delta p_{t,1} |\hat{y}_t - y_t^\star|.$$

Combining the two cases and using the fact that $\mathcal{I}_{0,1}$ and $\mathcal{I}_{0,0}$ are disjoint, we obtain

$$\begin{aligned}\bar{R}_0(T) &= \sum_{t\in\mathcal{I}_{0,1}} \bar{r}_t + \sum_{t\in\mathcal{I}_{0,0}} \bar{r}_t \\ &\le \frac{|\mathcal{I}_{0,0}|}{\gamma} + L_\Delta \sum_{t\in\mathcal{I}_0} p_{t,1}|\hat{y}_t - y_t^\star| \\ &\quad + \gamma L_\Delta^2 \sum_{t\in\mathcal{I}_{0,1}} p_{t,1}(\hat{y}_t - y_t^\star)^2.\end{aligned}$$

Since $\mathcal{I}_0 \subseteq [T]$, $\mathcal{I}_{0,1} \subseteq [T]$, and all terms are non-negative,

$$\begin{aligned}\bar{R}_0(T) &\le \frac{T}{\gamma} + L_\Delta \sum_{t=1}^{T} p_{t,1}|\hat{y}_t - y_t^\star| \\ &\quad + \gamma L_\Delta^2 \sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2.\end{aligned}$$

Taking expectation on both sides gives

$$\begin{aligned}\mathbb{E}\left[\bar{R}_0(T)\right] &\le \frac{T}{\gamma} + L_\Delta \mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}|\hat{y}_t - y_t^\star|\right] \\ &\quad + \gamma L_\Delta^2 \mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2\right].\end{aligned}$$

By the Cauchy–Schwarz inequality,

$$\begin{aligned}\sum_{t=1}^{T} p_{t,1}|\hat{y}_t - y_t^\star| &= \sum_{t=1}^{T} \sqrt{p_{t,1}}\sqrt{p_{t,1}}|\hat{y}_t - y_t^\star| \\ &\le \sqrt{\sum_{t=1}^{T} p_{t,1}} \sqrt{\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2} \\ &\le \sqrt{T\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2}.\end{aligned}$$

Taking expectation and applying Jensen's inequality,

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}|\hat{y}_t - y_t^\star|\right] &\le \sqrt{T\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2\right]} \\ &\le \sqrt{\frac{T\mathrm{Reg}_{\mathrm{KL}}(T)}{2}} \\ &= \frac{1}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)},\end{aligned}$$

where the last inequality follows from Lemma 2.

Also, by Lemma 2,

$$\mathbb{E}\left[\sum_{t=1}^{T} p_{t,1}(\hat{y}_t - y_t^\star)^2\right] \le \frac{1}{2}\mathrm{Reg}_{\mathrm{KL}}(T).$$

Substituting these bounds gives

$$\mathbb{E}\left[\bar{R}_0(T)\right] \le \frac{T}{\gamma} + \frac{L_\Delta}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)} + \frac{\gamma L_\Delta^2}{2}\mathrm{Reg}_{\mathrm{KL}}(T).$$

This completes the proof. □

Proof of Theorem 3. Suppose Assumptions 1 and 2 hold, and let $\mu = 2$. Let

$$L_{\mathrm{A}} := |\ell_{11} - \ell_{10}|, \qquad L_{\Delta} := |\ell_{01} - \ell_{11} + \ell_{10}|,$$

and

$$\Lambda := L_{\mathrm{A}}^2 + \frac{\ell_{01}^2}{\mu - 1}.$$

Then, for the choice

$$\gamma = \sqrt{\frac{5T}{\left(\frac{\Lambda}{8} + \frac{L_{\Delta}^2}{2}\right) \mathrm{Reg}_{\mathrm{KL}}(T)}}$$

the expected cumulative regret of LogCB-AT satisfies

$$\mathbb{E}[R(T)] \leq 2\sqrt{5T\left(\frac{\Lambda}{8} + \frac{L_{\Delta}^2}{2}\right) \mathrm{Reg}_{\mathrm{KL}}(T)} + \frac{L_{\Delta}}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)}.$$

Consequently,

$$\mathbb{E}[R(T)] = O\left(\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)}\right).$$

From the regret decomposition,

$$\mathbb{E}[R(T)] = \mathbb{E}\left[\bar{R}_1(T) + \bar{R}_0(T)\right].$$

By Lemma 3,

$$\mathbb{E}\left[\bar{R}_1(T)\right] \leq \frac{4T}{\gamma} + \frac{\gamma\Lambda}{8}\mathrm{Reg}_{\mathrm{KL}}(T).$$

By Lemma 4,

$$\mathbb{E}\left[\bar{R}_0(T)\right] \leq \frac{T}{\gamma} + \frac{L_{\Delta}}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)} + \frac{\gamma L_{\Delta}^2}{2}\mathrm{Reg}_{\mathrm{KL}}(T).$$

Combining these two bounds yields

$$\mathbb{E}[R(T)] \leq \frac{5T}{\gamma} + \gamma\left(\frac{\Lambda}{8} + \frac{L_{\Delta}^2}{2}\right)\mathrm{Reg}_{\mathrm{KL}}(T) + \frac{L_{\Delta}}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)}.$$

We choose γ to balance the first two terms:

$$\frac{5T}{\gamma} = \gamma\left(\frac{\Lambda}{8} + \frac{L_{\Delta}^2}{2}\right)\mathrm{Reg}_{\mathrm{KL}}(T).$$

This gives

$$\gamma = \sqrt{\frac{5T}{\left(\frac{\Lambda}{8} + \frac{L_{\Delta}^2}{2}\right)\mathrm{Reg}_{\mathrm{KL}}(T)}}.$$

Substituting this choice of γ gives

$$\mathbb{E}[R(T)] \leq 2\sqrt{5T\left(\frac{\Lambda}{8} + \frac{L_{\Delta}^2}{2}\right)\mathrm{Reg}_{\mathrm{KL}}(T)} + \frac{L_{\Delta}}{\sqrt{2}}\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)}.$$

Since Λ and L_{Δ} depend only on the fixed loss parameters, we conclude that

$$\mathbb{E}[R(T)] = O\left(\sqrt{T\mathrm{Reg}_{\mathrm{KL}}(T)}\right).$$

□

B Regret Bound Comparison: LogCB-AT vs. SquareCB.G-AT

We refer to the apple-tasting specialization of SquareCB.G as SquareCB.G-AT. For a comparison under the same shared binary outcome structure, suppose that its two action-loss predictions are constructed from a single estimate $\widehat{y}_t$:

$$\widehat{f}_{t,a} = \psi_a(\widehat{y}_t), \qquad f_a^\star(x_t) = \psi_a(y_t^\star), \qquad a \in \{0,1\}.$$

SquareCB.G-AT chooses Accept with probability

$$p_{t,1} = \begin{cases} 1, & \widehat{f}_{t,1} \le \widehat{f}_{t,0}, \\ \dfrac{2}{4 + \gamma_{\mathrm{Sq}}(\widehat{f}_{t,1} - \widehat{f}_{t,0})}, & \widehat{f}_{t,1} > \widehat{f}_{t,0}, \end{cases} \qquad p_{t,0} = 1 - p_{t,1}.$$

The bound $\mathrm{dec}_{\gamma_{\mathrm{Sq}}} \le 6/\gamma_{\mathrm{Sq}}$ for the apple tasting feedback graph gives

$$\mathbb{E}[R(T)] \le \frac{6T}{\gamma_{\mathrm{Sq}}} + \frac{\gamma_{\mathrm{Sq}}}{4}\,\mathrm{Reg}_{\mathrm{Sq}}(T).$$

Under the shared binary outcome parameterization,

$$\begin{aligned} \sum_{a=0}^{1} \big(\widehat{f}_{t,a} - f_a^\star(x_t)\big)^2 &= \big[\ell_{01}^2 + (\ell_{10} - \ell_{11})^2\big](\widehat{y}_t - y_t^\star)^2 \\ &= \Lambda(\widehat{y}_t - y_t^\star)^2, \end{aligned}$$

where $\Lambda := \ell_{01}^2 + (\ell_{10} - \ell_{11})^2$. Let $\mathrm{Reg}_{\mathrm{Sq}}^{y}(T)$ be the square-loss oracle regret bound for predicting the shared binary outcome on Accept rounds. Since observing the label reveals both action losses, the corresponding action-loss oracle regret is bounded by $\Lambda\,\mathrm{Reg}_{\mathrm{Sq}}^{y}(T)$. Optimizing the resulting bound over γ_{Sq} yields

$$\mathbb{E}[R(T)] \le C_{\mathrm{Sq}}\sqrt{T\,\mathrm{Reg}_{\mathrm{Sq}}^{y}(T)}, \qquad C_{\mathrm{Sq}} := \sqrt{6\Lambda},$$

provided the optimizing choice of γ_{Sq} satisfies the SquareCB.G requirement $\gamma_{\mathrm{Sq}} \ge 4$.

For LogCB-AT, the bound derived above, with $\mu = 2$, is

$$\mathbb{E}[R(T)] \le C_{\mathrm{Log}}\sqrt{T\,\mathrm{Reg}_{\mathrm{KL}}(T)}, \qquad C_{\mathrm{Log}} := \sqrt{\frac{5}{2}\Lambda + 10L_\Delta^2} + \frac{L_\Delta}{\sqrt{2}},$$

where $L_\Delta := |\ell_{01} - \ell_{11} + \ell_{10}|$. To compare the dependence of the two coefficients on the loss matrix, set $\rho := L_\Delta^2/\Lambda$. Then

$$\frac{C_{\mathrm{Sq}}}{\sqrt{\Lambda}} = \sqrt{6}, \qquad \frac{C_{\mathrm{Log}}}{\sqrt{\Lambda}} = \sqrt{\frac{5}{2} + 10\rho} + \sqrt{\frac{\rho}{2}}.$$

The coefficients are equal at

$$\rho^\star = \left(\frac{\sqrt{145} - 2\sqrt{3}}{19}\right)^2 \approx 0.204.$$

Thus $C_{\mathrm{Log}} < C_{\mathrm{Sq}}$ when $\rho < \rho^\star$, $C_{\mathrm{Log}} = C_{\mathrm{Sq}}$ when $\rho = \rho^\star$, and $C_{\mathrm{Log}} > C_{\mathrm{Sq}}$ when $\rho > \rho^\star$. The oracle regrets also matter: for a fixed loss matrix, the displayed LogCB-AT bound is smaller when

$$C_{\mathrm{Log}}^2\,\mathrm{Reg}_{\mathrm{KL}}(T) < C_{\mathrm{Sq}}^2\,\mathrm{Reg}_{\mathrm{Sq}}^{y}(T),$$

and the SquareCB.G-AT bound is smaller when the inequality is reversed. Hence a larger loss-dependent coefficient can be offset by a smaller oracle regret.

C LogCBPSide-AT Regret Analysis

In this section we provide the detailed proof for Theorem 1.

Our reward model is,

$$\mathbb{E}\left[y_t \mid x_t\right] = \sigma\left(x_t^T \cdot \theta^*\right)$$

Following the work of Filippi et al. (2010), we use the ℓ_2-regularized maximum quasi-likelihood estimator defined in Equation 2. When the minimizer lies in the interior of Θ, its first-order condition is

$$\sum_{s=1}^{t} \mathbb{1}\{a_s = 1\}\left(y_s - \sigma(x_s^T \hat{\theta}_t)\right) x_s - \lambda \hat{\theta}_t = \mathbf{0}.$$

Writing $\varepsilon_s := y_s - \sigma(x_s^T \theta^*)$, the sequence $\mathbb{1}\{a_s = 1\}\varepsilon_s x_s$ is conditionally mean-zero given the context and history, because the acceptance decision is made before observing y_s. The regularized confidence bound below follows by combining the corresponding vector-martingale concentration argument with the ridge-bias bound $\|\lambda\theta^*\|_{V_{t-1}^{-1}} \le \sqrt{\lambda} S$.

Proposition 1 (High probability confidence bound adapted from Proposition 1 of Filippi et al. (2010))**.** *Fix $\lambda > 0$ and $S > 0$, assume $\theta^* \in \Theta$, and take δ_t, N_{t-1} such that $0 < \delta_t < \min\{1, d/e\}$ and $1 \le N_{t-1} \le T$. Let*

$$V_{t-1} := \lambda I + \sum_{s=1}^{t-1} \mathbb{1}\{a_s = 1\} x_s x_s^T,$$

$$\beta_{t-1}^{x_t}(\delta_t) := \frac{k_\sigma}{c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \left(2R_{\max}\sqrt{q_{N_{t-1}}(\delta_t)} + \sqrt{\lambda} S\right).$$

Then, with probability at least $1 - \delta_t$, it holds that

$$|\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| \le \beta_{t-1}^{x_t}(\delta_t).$$

where,

- $R_{\max} := \max_t y_t = 1$.
- k_σ := *Lipschitz constant of* σ.

$$\begin{aligned} \sigma'(z) &= \sigma(z)(1 - \sigma(z)) \\ &\le \frac{1}{4} \\ k_\sigma &= \frac{1}{4}. \end{aligned}$$

- $C_{\max} := \sup_{x \in \mathcal{X}, \theta \in \Theta} |x^T \theta|$.
- $c_\sigma := \inf_{x \in \mathcal{X}, \theta \in \Theta} \sigma'(x^T \theta) > 0$.

$$c_\sigma = \frac{e^{C_{\max}}}{\left(1 + e^{C_{\max}}\right)^2}.$$

For the choice $\delta_t = 1/t$, define the uniform coefficient

$$h(T) := \frac{k_\sigma}{c_\sigma}\left[2R_{\max}\sqrt{\left(3 + 2\log\left(1 + \frac{2}{\lambda}\right)\right) 2d\log(T+1)\log(dT)} + \sqrt{\lambda} S\right]. \tag{9}$$

Then $\beta_{t-1}^{x_t}(\delta_t) \leq h(T)\|x_t\|_{V_{t-1}^{-1}}$ for all confidence-bound rounds. For fixed λ and S, the additional ridge-bias term changes constants but not the stated asymptotic orders in T.

Furthermore, we introduce 2 more Lemmas necessary for the regret analysis.

The Elliptical Potential Lemma (EPL),

Lemma 5 (Elliptical potential lemma adapted from Proposition 2 of Abeille & Lazaric (2017))**.** *Let $x_1, x_2, ..., x_t \in \mathbb{R}^d$ be a sequence of vectors with $\|x_s\|_2 \leq 1, \forall s \in [t]$. Let $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T$ for some $\lambda > 0$. Then,*

$$\sum_{s=1}^{t} \|x_s\|^2_{V_{s-1}^{-1}} \leq 2d \log(1 + \frac{t}{d\lambda}).$$

Adapting this to our case:

Let $x_1 \cdot \mathbb{1}\{a_1 = 1\}, x_2 \cdot \mathbb{1}\{a_2 = 1\}, ..., x_t \cdot \mathbb{1}\{a_t = 1\} \in \mathbb{R}^d$ for some $\{a_s\}_{s=1}^t \in \{0, 1\}$. Let $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T \mathbb{1}\{a_s = 1\}$ for some $\lambda > 0$. Then,

$$\sum_{s=1}^{t} \|x_s\|^2_{V_{s-1}^{-1}} \mathbb{1}\{a_s = 1\} \leq 2d \log(1 + \frac{t}{d\lambda}).$$

and Elliptical Potential Count Lemma (EPC)

Lemma 6 (Elliptical potential count lemma adapted from Lemma C.2 of Jun & Kim (2024))**.** *Let $x_1, x_2, ..., x_t \in \mathbb{R}^d$ be a sequence of vectors with $\|x_s\|_2 \leq 1, \forall s \in [t]$. Let $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T$ for some $\lambda > 0$. Let $J = \{s \in [t] : \|x_s\|^2_{V_{s-1}^{-1}} \geq L^2\}$ for some $L^2 \leq 1$. Then,*

$$|J| \leq 3\frac{d}{L^2} \ln\left(1 + \frac{2}{L^2\lambda}\right).$$

Adapting this to our case:

Let $x_1 \cdot \mathbb{1}\{a_1 = 1\}, x_2 \cdot \mathbb{1}\{a_2 = 1\}, ..., x_t \cdot \mathbb{1}\{a_t = 1\} \in \mathbb{R}^d$ for some $\{a_s\}_{s=1}^t \in \{0, 1\}$. Let $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T \mathbb{1}\{a_s = 1\}$ for some $\lambda > 0$. Let $J = \{s \in [t] : \|x_s \mathbb{1}\{a_s = 1\}\|^2_{V_{s-1}^{-1}} \geq L^2\}$ for some $L^2 \leq 1$. Then,

$$|J| \leq 3\frac{d}{L^2} \ln\left(1 + \frac{2}{L^2\lambda}\right).$$

C.1 Minimax regret analysis

Theorem 1 (LogCBPSide-AT minimax regret bound)**.** *For $\delta_t = \frac{1}{t}$, fixed $\lambda > 0$ and $S > 0$, $\theta^* \in \Theta$, $\sup_{x \in \mathcal{X}} \|x\|_2 \leq 1$, and $C_{\max} := \sup_{x \in \mathcal{X}, \theta \in \Theta} |x^T\theta| < \infty$, the expected cumulative regret of the LogCBPSide-AT algorithm is bounded as:*

$$\mathbb{E}\left[R(T)\right] \leq O\left(\frac{\left(1 + e^{C_{\max}}\right)^2}{e^{C_{\max}}} d\sqrt{T} \log^{\frac{3}{2}}(T)\right).$$

(Ignoring logarithmic terms in d)

Proof. Throughout the analysis, we use $\hat{y}_t = \sigma(x_t^\top \hat{\theta}_{t-1})$ and $y_t^* = \sigma(x_t^\top \theta^*)$ to denote the estimated and true conditional means, respectively.

Let us define the good event $\mathcal{G}_t$ as,

$$\mathcal{G}_t = \left\{|\sigma(x_t^T\theta^*) - \sigma(x_t^T\hat{\theta}_{t-1})| \leq \beta_{t-1}^{x_t}(\delta_t).\right\}$$

Instantaneous regret,

$$r_t = \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)$$

Let $\mathcal{F}_{t-1} = \sigma((x_1, a_1, y_1), \ldots, (x_{t-1}, a_{t-1}, y_{t-1}), x_t)$ be the filtration representing the history up to round t. Furthermore, use the shorthand $\mathbb{E}_{t-1}[\cdot]$ to denote $\mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$.

$$\mathbb{E}_{t-1}\left[r_t\right] = \psi_{a_t}(y_t^*) - \psi_{a_t^*}(y_t^*)$$

Cumulative regret,

$$\mathrm{Reg}_T := \mathbb{E}\left[R(T)\right] = \mathbb{E}\left[\sum_{t=1}^{T} r_t\right]$$

$$\mathrm{Reg}_T = \underbrace{\mathbb{E}\left[\sum_{t=1}^{T} r_t \mathbb{1}\left\{\mathcal{G}_t\right\}\right]}_{F_1} + \underbrace{\mathbb{E}\left[\sum_{t=1}^{T} r_t \mathbb{1}\left\{\bar{\mathcal{G}}_t\right\}\right]}_{F_2}$$

First, let us bound F_2

$$\begin{aligned}
F_2 &= \mathbb{E}\left[\sum_{t=1}^{T} r_t \mathbb{1}\left\{\bar{\mathcal{G}}_t\right\}\right] \\
&= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^{T} \mathbb{P}\left(\bar{\mathcal{G}}_t\right) \\
&= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^{T} \delta_t \\
&= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^{T} \frac{1}{t} && \text{(Proposition 1, set } \delta_t = \tfrac{1}{t}\text{)} \\
&\lessapprox \max(\ell_{10}, \ell_{01} - \ell_{11}) \log T \\
&= O(\log T)
\end{aligned}$$

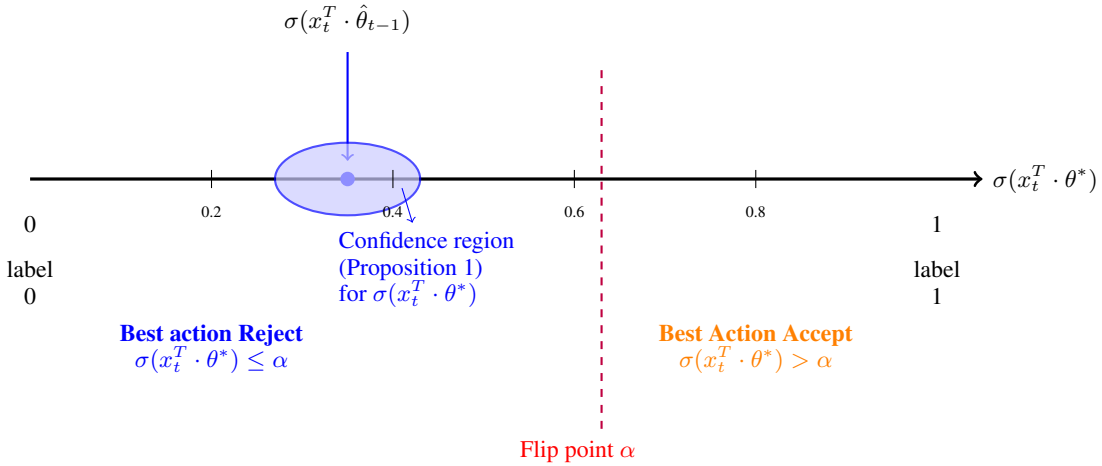
Now let us bound F_1,

$$\begin{aligned}
F_1 &= \sum_{t=1}^{T} \mathbb{E}\left[r_t \mathbb{1}\left\{\mathcal{G}_t\right\}\right] \\
&= \underbrace{\sum_{t=1}^{T} \mathbb{E}\left[r_t \mathbb{1}\left\{\mathcal{G}_t\right\} \mathbb{1}\left\{\sigma(x_t^T \hat{\theta}_{t-1}) \le \alpha - \beta_{t-1}^{x_t}(\delta_t)\right\}\right]}_{F_{11}} + \underbrace{\sum_{t=1}^{T} \mathbb{E}\left[r_t \mathbb{1}\left\{\mathcal{G}_t\right\} \mathbb{1}\left\{\sigma(x_t^T \hat{\theta}_{t-1}) \ge \alpha + \beta_{t-1}^{x_t}(\delta_t)\right\}\right]}_{F_{12}} \\
&\quad + \underbrace{\sum_{t=1}^{T} \mathbb{E}\left[r_t \mathbb{1}\left\{\mathcal{G}_t\right\} \mathbb{1}\left\{\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T \hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)\right\}\right]}_{F_{13}}.
\end{aligned}$$

Case F_{11}: $\hat{y}_t = \sigma(x_t^T \hat{\theta}_{t-1}) \le \alpha - \beta_{t-1}^{x_t}(\delta_t)$

First of all $|\hat{y}_t - \alpha| \geq \beta_{t-1}^{x_t}(\delta_t)$ and $\hat{y}_t \leq \alpha$, hence, $a_t = 0(\text{Reject})$ by the behavior of Algorithm 1

This is the case where, estimated distribution $\hat{y}_t$ is comfortably to the left of flip point, such that, based on Proposition 1, true mean of the distribution $\sigma(x_t^T\theta^*)$ itself will be on the left side of the flip point.



Since $\mathcal{G}_t$ happens,

$$
\begin{aligned}
|\sigma(x_t^T\theta^*) - \sigma(x_t^T\hat{\theta}_{t-1})| &\leq \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T\hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) \leq \sigma(x_t^T\theta^*) &\leq \sigma(x_t^T\hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T\theta^*) &\leq \alpha
\end{aligned}
$$

Hence, $a_t^* = 0$.

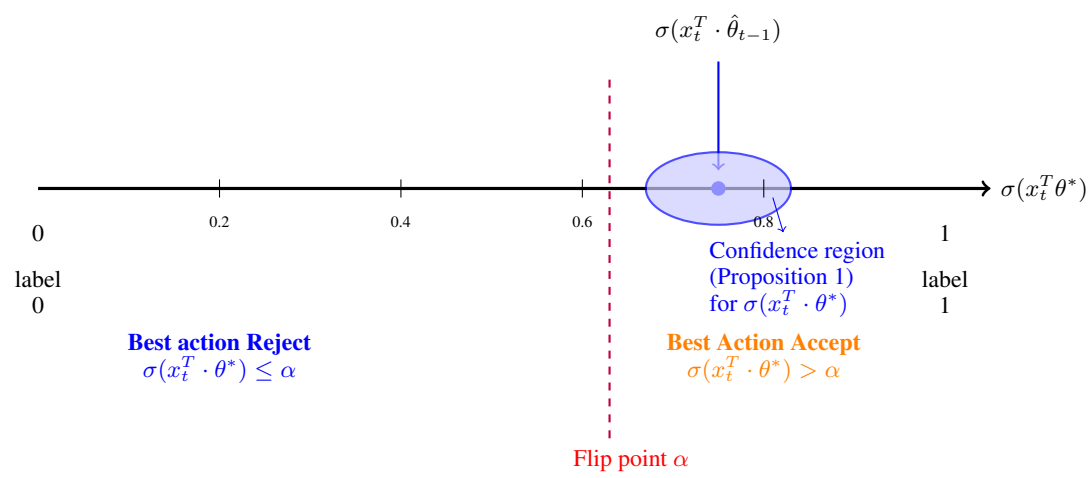
$$a_t = a_t^* = 0.$$

$$F_{11} = 0$$

Case F_{12}: $\hat{y}_t = \boldsymbol{\sigma(x_t^T\hat{\theta}_{t-1}) \geq \alpha + \beta_{t-1}^{x_t}(\delta_t)}$

First of all $|\hat{y}_t - \alpha| \geq \beta_{t-1}^{x_t}(\delta_t)$ and $\hat{y}_t \geq \alpha$, hence, $a_t = 1(\text{Accept})$

This is the case where, estimated distribution $\hat{y}_t$ is comfortably to the right of flip point, such that, based on Proposition 1, true mean of the distribution $\sigma(x_t^T\theta^*)$ itself will be on the right side of the flip point.



Since $\mathcal{G}_t$ happens,

$$|\sigma(x_t^T\theta^*) - \sigma(x_t^T\hat{\theta}_{t-1})| \leq \beta_{t-1}^{x_t}(\delta_t)$$
$$\sigma(x_t^T\hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) \leq \sigma(x_t^T\theta^*) \leq \sigma(x_t^T\hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t)$$
$$\sigma(x_t^T\theta^*) \geq \alpha$$

Hence $a_t^* = 1(\text{Accept})$.

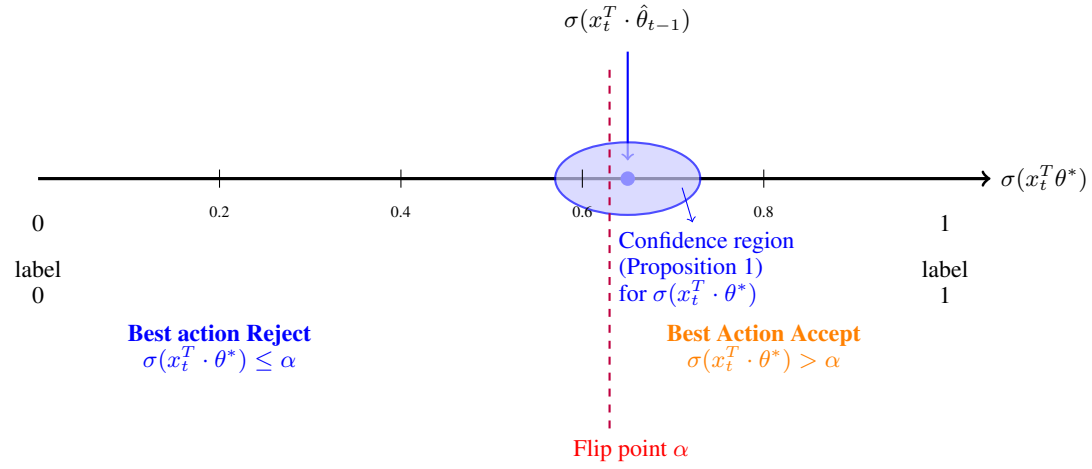
$$a_t = a_t^* = 1.$$

$$F_{12} = 0$$

Case F_{13}: $\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T\hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)$

First of all $|\hat{y}_t - \alpha| < \beta_{t-1}^{x_t}(\delta_t)$, hence, $a_t = 1(\text{Accept})$

This is an uncertain region which requires more exploration. However the nature of the confidence region $\beta_{t-1}^{x_t}(\delta_t)$ is such that it shrinks with more exploration (because its dependence on $\|x_t\|^2_{V_{t-1}^{-1}}$ which is a monotonically decreasing function). Hence the regret could be controlled using Elliptical potential lemma.



Since $\mathcal{G}_t$ happens,

$$
\begin{aligned}
|\sigma(x_t^T\theta^*) - \sigma(x_t^T\hat{\theta}_{t-1})| &\le \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T\hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) &\le \sigma(x_t^T\hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
\alpha - 2\beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) &\le \alpha + 2\beta_{t-1}^{x_t}(\delta_t) \\
-2\beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) - \alpha &\le 2\beta_{t-1}^{x_t}(\delta_t)
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}_{t-1}[r_t] &= \mathbb{E}_{t-1}\left[\psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)\right] \\
&= \mathbb{E}_{t-1}\left[\psi_1(y_t) - \psi_{a_t^*}(y_t)\right] \\
&\le \left|\mathbb{E}_{t-1}\left[\psi_1(y_t) - \psi_0(y_t)\right]\right| \\
&= \left|\mathbb{E}_{t-1}\left[\ell_{10} + y_t(\ell_{11} - \ell_{10}) - \ell_{01}y\right]\right| \\
&= \left|\left[\ell_{10} + \mathbb{E}_{t-1}[y_t](\ell_{11} - \ell_{10}) - \ell_{01}\,\mathbb{E}_{t-1}[y_t]\right]\right| \\
&= \left|\left[\ell_{10} + \sigma(x_t^T\theta^*)(\ell_{11} - \ell_{10}) - \ell_{01}\sigma(x_t^T\theta^*)\right]\right| \\
&= \left|\left[\left(\sigma(x_t^T\theta^*) - \alpha\right)(\ell_{11} - \ell_{10}) - \ell_{01}\left(\sigma(x_t^T\theta^*) - \alpha\right)\right]\right| \qquad \text{(definition of } \alpha\text{)} \\
&\le 2\beta_{t-1}^{x_t}(\delta_t)\max(\ell_{10}, \ell_{01} - \ell_{11})
\end{aligned}
$$

Let $L_{\max} := \max(\ell_{10}, \ell_{01} - \ell_{11})$. Using Equation 9,

$$
\begin{aligned}
F_{13} &= \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{\mathcal{G}_t\}\,\mathbb{1}\left\{\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T\hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)\right\}\right] \\
&\le \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{\mathcal{G}_t\}\,\mathbb{1}\{a_t = 1\}\right] \\
&\le 2L_{\max}\,\mathbb{E}\left[\sum_{t=1}^{T}\beta_{t-1}^{x_t}(\delta_t)\,\mathbb{1}\{a_t = 1\}\right] \\
&\le 2L_{\max}h(T)\,\mathbb{E}\left[\sum_{t=1}^{T}\|x_t\|_{V_{t-1}^{-1}}\,\mathbb{1}\{a_t = 1\}\right] \\
&\le 2L_{\max}h(T)\sqrt{T}\sqrt{\mathbb{E}\left[\sum_{t=1}^{T}\|x_t\|^2_{V_{t-1}^{-1}}\,\mathbb{1}\{a_t = 1\}\right]}.
\end{aligned}
$$

By applying Lemma 5,

$$
\begin{aligned}
F_{13} &\le 2L_{\max}h(T)\sqrt{2dT\log\left(1 + \frac{T}{d\lambda}\right)} \\
&= O\left(\frac{\left(1 + e^{C_{\max}}\right)^2}{e^{C_{\max}}}d\sqrt{T}\log^{3/2}(T)\right). \qquad \text{(for fixed } \lambda, S\text{; ignoring logarithmic terms in } d\text{)}
\end{aligned}
$$

Hence,

$$\begin{aligned}\mathrm{Reg}_T &\le 2L_{\max}h(T)\sqrt{2dT\log\left(1+\frac{T}{d\lambda}\right)}+\log T \\ &= O\left(\frac{\left(1+e^{C_{\max}}\right)^2}{e^{C_{\max}}}d\sqrt{T}\log^{3/2}(T)\right). \quad \text{(for fixed } \lambda, S\text{; ignoring logarithmic terms in } d\text{)}\end{aligned}$$

□

C.2 Instance dependent regret analysis

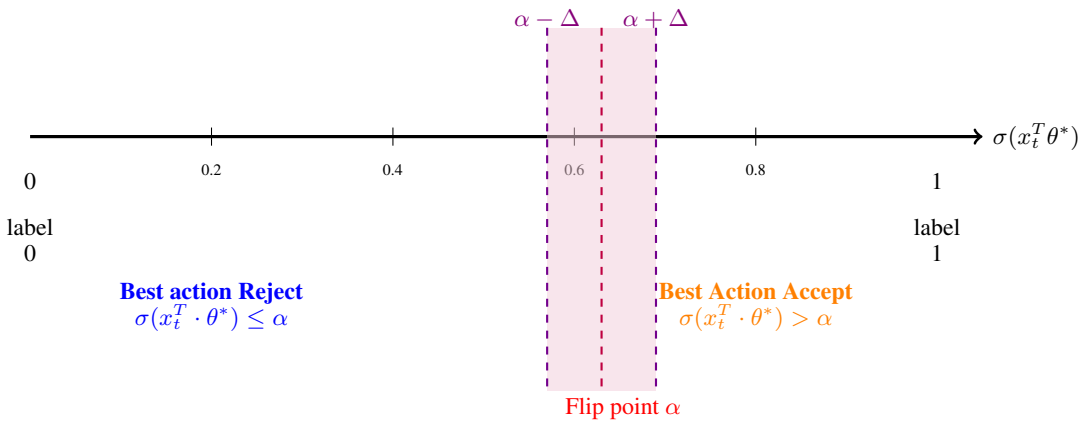
Theorem 2 (LogCBPSide-AT instance dependent regret bound)**.** *For $\delta_t = \frac{1}{t}$, fixed $\lambda > 0$ and $S > 0$, $\theta^* \in \Theta$, $\sup_{x\in\mathcal{X}}\|x\|_2 \le 1$, $C_{\max} := \sup_{x\in\mathcal{X},\theta\in\Theta}|x^T\theta| < \infty$, and $0 < \Delta \le \min_{x\in\mathcal{X}}|\alpha - \sigma(x^T\theta^*)|$, the expected cumulative regret of the LogCBPSide-AT algorithm is bounded as:*

$$\mathbb{E}\left[R(T)\right] \le O(\frac{1}{\Delta}d^2\log^2 T)$$

(ignoring doubly logarithmic factors and $\mathrm{polylog}(\frac{1}{\Delta})$)

Proof. Let define the problem dependent parameter $\Delta > 0$ as,

$$\Delta \le \min_{x\in\mathcal{X}}|\alpha - \sigma(x^T\cdot\theta^*)|$$



This defines how close $y_t^* = \sigma(x_t^T\cdot\theta^*)$ is to the decision boundary α. The closer, the problem instance become more difficult owing to the fact that we need more exploration to distinguish it. In the figure above, there are no instances of x_t falls into the shaded region.

With this additional assumption, we can refine our previous analysis.

F_2, F_{11}, F_{12} all go though for this case as well. We skip them and move to the analysis of F_{13}.

Case F_{13}: $\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T\hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)$

First of all $|\hat{y}_t - \alpha| < \beta_{t-1}^{x_t}(\delta_t)$, hence, $a_t = 1(\text{Accept})$

Since $\mathcal{G}_t$ happens,

$$
\begin{aligned}
|\sigma(x_t^T\theta^*) - \sigma(x_t^T\hat{\theta}_{t-1})| &\le \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T\hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) &\le \sigma(x_t^T\hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
\alpha - 2\beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) &\le \alpha + 2\beta_{t-1}^{x_t}(\delta_t) \\
-2\beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) - \alpha &\le 2\beta_{t-1}^{x_t}(\delta_t).
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}_{t-1}[r_t] &= \mathbb{E}_{t-1}\left[\psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)\right] \\
&= \mathbb{E}_{t-1}\left[\psi_1(y_t) - \psi_{a_t^*}(y_t)\right] \\
&\le \left|\mathbb{E}_{t-1}\left[\psi_1(y_t) - \psi_0(y_t)\right]\right| \\
&= \left|\mathbb{E}_{t-1}\left[\ell_{10} + y_t(\ell_{11} - \ell_{10}) - \ell_{01}y\right]\right| \\
&= \left|\left[\ell_{10} + \mathbb{E}_{t-1}[y_t](\ell_{11} - \ell_{10}) - \ell_{01}\mathbb{E}_{t-1}[y_t]\right]\right| \\
&= \left|\left[\ell_{10} + \sigma(x_t^T\theta^*)(\ell_{11} - \ell_{10}) - \ell_{01}\sigma(x_t^T\theta^*)\right]\right| \\
&= \left|\left[\left(\sigma(x_t^T\theta^*) - \alpha\right)(\ell_{11} - \ell_{10}) - \ell_{01}\left(\sigma(x_t^T\theta^*) - \alpha\right)\right]\right| \qquad \text{(definition of } \alpha\text{)} \\
&\le 2\beta_{t-1}^{x_t}(\delta_t)\max(\ell_{10}, \ell_{01} - \ell_{11})
\end{aligned}
$$

$$
\begin{aligned}
F_{13} &= \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{\mathcal{G}_t\}\,\mathbb{1}\left\{\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T\hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)\right\}\right] \\
&\le \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{-2\beta_{t-1}^{x_t}(\delta_t) \le \sigma(x_t^T\theta^*) - \alpha \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\right] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\right] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\mathbb{1}\left\{\beta_{t-1}^{x_t}(\delta_t) < \frac{\Delta}{2}\right\}\right] \\
&\quad + \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\mathbb{1}\left\{\beta_{t-1}^{x_t}(\delta_t) \ge \frac{\Delta}{2}\right\}\right] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| < \Delta\right\}\right] \\
&\quad + \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\mathbb{1}\left\{\beta_{t-1}^{x_t}(\delta_t) \ge \frac{\Delta}{2}\right\}\right] \\
&= 0 \qquad \text{(By definition of } \Delta\text{)} \\
&\quad + \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\mathbb{1}\left\{\beta_{t-1}^{x_t}(\delta_t) \ge \frac{\Delta}{2}\right\}\right] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[r_t\,\mathbb{1}\{a_t = 1\}\,\mathbb{1}\left\{|\sigma(x_t^T\theta^*) - \alpha| \le 2\beta_{t-1}^{x_t}(\delta_t)\right\}\mathbb{1}\left\{\beta_{t-1}^{x_t}(\delta_t) \ge \frac{\Delta}{2}\right\}\right] \\
&\le 2\max(\ell_{10}, \ell_{01} - \ell_{11})\sum_{t=1}^{T}\mathbb{E}\left[\mathbb{1}\{a_t = 1\}\,\beta_{t-1}^{x_t}(\delta_t)\,\mathbb{1}\left\{\beta_{t-1}^{x_t}(\delta_t) \ge \frac{\Delta}{2}\right\}\right]
\end{aligned}
$$

From Proposition 1,

$$\beta_{t-1}^{x_t}(\delta_t) \le h(T)\|x_t\|_{V_{t-1}^{-1}},$$
$$\beta_{t-1}^{x_t}(\delta_t) \ge \frac{\Delta}{2} \implies \|x_t\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)},$$

$$\begin{aligned}
F_{13} &\le 2\max(\ell_{10}, \ell_{01}-\ell_{11}) \sum_{t=1}^{T} \mathbb{E}\left[\mathbb{1}\{a_t=1\}\,\beta_{t-1}^{x_t}(\delta_t)\,\mathbb{1}\left\{\|x_t\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\right\}\right] \\
&\le 2\max(\ell_{10}, \ell_{01}-\ell_{11}) h(T) \sum_{t=1}^{T} \mathbb{E}\left[\mathbb{1}\{a_t=1\}\,\|x_t\|_{V_{t-1}^{-1}}\,\mathbb{1}\left\{\|x_t\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\right\}\right]
\end{aligned}$$

By applying peeling method,

$$\begin{aligned}
F_{13} &\le 2\max(\ell_{10}, \ell_{01}-\ell_{11}) h(T) \sum_{t=1}^{T} \mathbb{E}\left[\mathbb{1}\{a_t=1\}\,\|x_t\|_{V_{t-1}^{-1}}\,\mathbb{1}\left\{\|x_t\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\right\}\right] \\
&= 2\max(\ell_{10}, \ell_{01}-\ell_{11}) h(T) \sum_{t=1}^{T}\sum_{\ell=1}^{\infty} \mathbb{E}\left[\mathbb{1}\{a_t=1\}\,\|x_t\|_{V_{t-1}^{-1}}\,\mathbb{1}\left\{2^{\ell+1}\frac{\Delta}{2h(T)} \ge \|x_t\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\cdot 2^{\ell}\right\}\right] \\
&\le 2\max(\ell_{10}, \ell_{01}-\ell_{11}) h(T) \sum_{t=1}^{T}\sum_{\ell=1}^{\infty} \mathbb{E}\left[\mathbb{1}\{a_t=1\}\,2^{\ell+1}\frac{\Delta}{2h(T)}\,\mathbb{1}\left\{2^{\ell+1}\frac{\Delta}{2h(T)} \ge \|x_t\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\cdot 2^{\ell}\right\}\right] \\
&\le \max(\ell_{10}, \ell_{01}-\ell_{11}) \sum_{t=1}^{T}\sum_{\ell=1}^{\infty} \mathbb{E}\left[2^{\ell+1}\Delta\,\mathbb{1}\left\{2^{\ell+1}\frac{\Delta}{2h(T)} \ge \|x_t\,\mathbb{1}\{a_t=1\}\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\cdot 2^{\ell}\right\}\right] \\
&= \max(\ell_{10}, \ell_{01}-\ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1}\Delta \sum_{t=1}^{T} \mathbb{E}\left[\mathbb{1}\left\{2^{\ell+1}\frac{\Delta}{2h(T)} \ge \|x_t\,\mathbb{1}\{a_t=1\}\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\cdot 2^{\ell}\right\}\right] \\
&\le \max(\ell_{10}, \ell_{01}-\ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1}\Delta \sum_{t=1}^{T} \mathbb{E}\left[\mathbb{1}\left\{\|x_t\,\mathbb{1}\{a_t=1\}\|_{V_{t-1}^{-1}} \ge \frac{\Delta}{2h(T)}\cdot 2^{\ell}\right\}\right] \\
&= \max(\ell_{10}, \ell_{01}-\ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1}\Delta\,\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{1}\left\{\|x_t\,\mathbb{1}\{a_t=1\}\|^2_{V_{t-1}^{-1}} \ge \frac{\Delta^2}{4h^2(T)}\cdot 2^{2\ell}\right\}\right] \\
&\le \max(\ell_{10}, \ell_{01}-\ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1}\Delta\,\mathbb{E}\left[\frac{12dh^2(T)}{2^{2\ell}\Delta^2}\log\left(1+\frac{8h^2(T)}{\lambda 2^{2\ell}\Delta^2}\right)\right] \qquad \text{(Lemma 6 (EPC))} \\
&\le \frac{24dh^2(T)}{\Delta}\max(\ell_{10}, \ell_{01}-\ell_{11})\log\left(1+\frac{2h^2(T)}{\lambda\Delta^2}\right)\sum_{\ell=1}^{\infty}2^{-\ell} \\
&\le \frac{24dh^2(T)}{\Delta}\max(\ell_{10}, \ell_{01}-\ell_{11})\log\left(1+\frac{2h^2(T)}{\lambda\Delta^2}\right)
\end{aligned}$$

Hence,

$$\begin{aligned}
\mathrm{Reg}_T &\le \frac{24dh^2(T)}{\Delta}\max(\ell_{10}, \ell_{01}-\ell_{11})\log\left(1+\frac{2h^2(T)}{\lambda\Delta^2}\right) + O(\log T) \\
&= O\left(\frac{1}{\Delta}d^2\log^2 T\right).
\end{aligned}$$

(for fixed λ, S; ignoring doubly logarithmic factors and $\mathrm{polylog}(1/\Delta)$)

□

D EXPERIMENT DETAILS

We present the detailed procedure of AppleTeA in Algorithm 3.

Algorithm 3 AppleTeA

Input: $\mathcal{M}_s$, $\mathcal{M}_\ell$, n, ϕ, α, γ, λ, S, μ, algorithm Alg
if Alg = "LogCB-AT" **then**
  Instantiate $\mathcal{A} = \text{LogCB-AT}(\gamma, \mu)$
else
  Instantiate $\mathcal{A} = \text{LogCBPSide-AT}(\lambda, S, \alpha)$
end if
for $t = 1, \ldots, n$ **do**
  Observe query q_t
  Run SLM: $r_{s,t} = \mathcal{M}_s(q_t)$
  Extract the SLM answer: $\hat{y}_{w,t} = \text{answer}(r_{s,t})$
  Compute context: $x_t = \phi(q_t, r_{s,t})$
  Run one step of $\mathcal{A}$ and obtain action a_t
  if $a_t = 0$ (Reject) **then**
    Output $\hat{y}_{w,t}$
  else
    Run the strong LLM: $r_{\ell,t} = \mathcal{M}_\ell(q_t)$
    Extract the LLM answer: $\hat{y}_{s,t} = \text{answer}(r_{\ell,t})$
    Set $y_t = \mathbb{1}\{\hat{y}_{s,t} \neq \hat{y}_{w,t}\}$
    Update $\mathcal{A}$ with (x_t, y_t)
    Output $\hat{y}_{s,t}$
  end if
end for

Context Construction. We construct the context vector $x_t \in \mathbb{R}^d$ using the SLM hidden representation and prediction uncertainty. Specifically, we extract the hidden state at the token position immediately preceding the answer token and apply dimensionality reduction.

For the uncertainty component, we consider two scores. The maximum-probability-based uncertainty score is defined as

$$p_t^{\max} = 1 - \max_{v \in \mathcal{V}} \rho(\text{logit}_{t,v}),$$

where ρ denotes the softmax function. The margin-based uncertainty score is defined as

$$p_t^{\text{margin}} = 1 - \left(\text{logit}_{t,v^{(1)}} - \text{logit}_{t,v^{(2)}}\right),$$

where $v^{(1)}$ and $v^{(2)}$ denote the top two candidate tokens. The context vector is constructed by concatenating the dimensionally reduced hidden representation with the uncertainty score.